PROBABILITY THEORY
MTM2592-Gr:2

FIRST WEEK

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EVALUATION SYSTEM

Midterm → 60 %

Final → 40 %

Percentage of In-Term Studies: 60%

Percentage of Final Examination:40%

Course Learning Outcomes

- Having knowledge about the basic concepts of Probability Theory.
- Having ability to model probability problems.
- Solving these models.
- Interpreting results of problems that are modeled and solved.
- Take an active role in interdisciplinary team work.

Recommended Sources

- Probability and Statistics for Engineering and Scientists, R.E. Walpole, R.H. Myers, S.L. Myers, K.Ye, Prentice Hall, 2007.
- John E. Freund's Mathematical Statistics, Miller I., Miller M, Prentice Hall, 6th edition, 1999.
- Olasılık, Doç. Dr. Cevdet Cerit, Y.Doç.Dr. Muserref Yuksel, Beta Yayınevi, 2004.
- Prof. Dr. Ibrahim Sezginman, "Olasılık Teori ve Problemleri", YUVAK Yay., 2001.
- P.L. Meyer, "Introductory Probability and Statistical Applications", Addison-Wesley Pub., 1978.
- Prof. Dr. Aziz Bener, "Matematik İstatistik (Olasılık Hesabı)", C.I., YTÜVAK Yay., 2002.
- Prof.Dr. Fikri Akdeniz, "Olasılık ve İstatistik", Nobel Kitabevi, 2007.

Weekly Subjects

Week	Subjects
1	Set Theory, Counting Techniques, Permutation, Permutation with Repetition, Combination, Combination with Repetition
2	Binomial Expansion, Multinomial Expansion, Tree diagrams, Random Experiment, sample space, Set Algebra of Events, Probability Space, Axiom of Probability
3	Conditional Probability, Independence, Bayes Theorem, Random Variable, Functions of Probability
4	Distribution, Functions of Density, Expected Value, Chebycheff’s Inequality
5	Moment Generation Function, Mixed Random Variables
6	Multivariable Distribution, Joint Probability Distributions
7	Marginal distributions, conditional distributions, independency
8	Midterm 1
9	Conditional expected value and conditional variance, Means and Variance of Linear Combinations of Random Variables
10	Some Discrete Probability Distributions: Discrete Uniform Distribution, Bernoulli Distribution, Binomial Distribution, Geometric Distribution,
11	Negative Binomial Distribution, Hypergeometric Distribution, Multinomial distribution, Multivariate hypergeometric distribution
12	Poisson Distribution, Some Continuous Probability Distributions: Continuous Uniform Distribution, Gama Distribution
13	Chi-Squared Distributions, Beta Distribution, Exponential Distribution
14	Normal Distribution,Applications of Normal Distribution

THE BASICS OF SET THEORY

A set is a well-defined group of objects or ***elements***

- {1, 2, 3}
- $\{x \in \mathbf{R} \mid x^2 > 5\}$
- S = {Tom, Sue, Jim}

We use capital letters to denote sets

- $S=\{a, b, c\}$ refers to the set

  whose elements are a, b and c.
- $a \in S$ means "a is an element of set S

  or (a belongs to the set S)".
- $d \notin S$ means "d is *not* an element of set S

  or (d does not belongs to the set S)".
- $\{x \in S \mid P(x)\}$ is the set of all those x from S such that P(x) is true. *E.g.*, $T=\{x \in \mathbf{Z} \mid 0<x<10\}$.
- *Notes:*

  Sets can themselves be elements of other sets, *e.g.*,

  S={ {Mary, John}, {Tim, Ann}, …}

Subsets

- If A and B are sets, A is called a *subset* of B, denoted $A \subseteq B$, provided every element of A is an element of B.
- So, $A \subseteq B$ means $\forall x$, if $x \in A$, then $x \in B$.
- We also say, "A *is contained in* B" or "B *contains* A" to show this relationship.
- Equivalently, we denote $A \not\subseteq B$ provided $\exists x \in A$ and $x \notin B$.

Examples of Subsets

- If $A = \{1, 2, 3\}$ and $B = \{0, 1, 2, 3, 4\}$, then clearly $A \subseteq B$.
- $\{\{1\}, \{2\}\} \subseteq \{\{1\}, \{2\}, \{1,2\}\}$.
- We denote *interval* subsets of **R** as $[a, b) = \{x \in \mathbf{R} \mid a \leq x < b\}$. So $[2, 5) \subseteq [0,5]$.

Proper subset definition

A proper subset of a set A is a subset of A that is not equal to A. In other words, if B is a proper subset of A, then all elements of B are in A but A contains at least one element that is not in B.

For example, if $A = \{1, 3, 5\}$ then $B = \{1, 5\}$ is a proper subset of A. The set $C = \{1, 3, 5\}$ is a subset of A, but it is not a proper subset of A since $C = A$. The set $D = \{1, 4\}$ is not even a subset of A, since 4 is not an element of A.

$C \subseteq A$ a subset,

$B \subset A$ a proper subset

The set that contains no elements is called empty set and is denoted by $\varnothing$ or {}.

Theorem

- $\phi \subseteq A$. If A is not empty, then $\phi \subset A$.

Set Equality

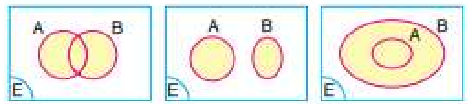
- We say sets A and B are *equal* (A = B) if every element of A is in B and every element of B is in A.
- Thus, A = B means $A \subseteq B$ and $B \subseteq A$.
- For example $\{1, 2, 3\} = \{1, 2, 3\}$, but $A = \{1, 2, 3\} \neq \{1, 2, 3, 4\} = B$, since $4 \in B$ but $4 \notin A$.
- Also, $[a, b) \neq [a, b]$ since b is only in $[a, b]$.

Operations on Sets

or E

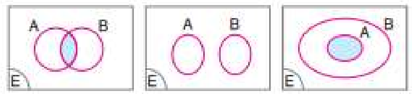
- Given sets A and B, which are subsets of a universal set, U, we define the following:
- **Union:** The set consisting of two sets is their union

  $A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}.$



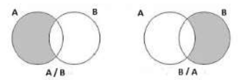
- **Intersection:** The set consisting of common elements of two sets is their intersection.

  $A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}.$



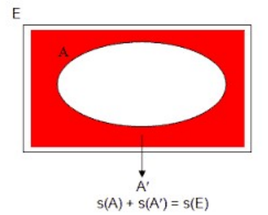
- **Difference:** Set difference of A and B denoted by A-B (or A/B), is the set of all members (elements) of A that are not members of B.

  $A - B = \{x \in U \mid x \in A \text{ and } x \notin B\}.$



- **Complement:** The set difference U-A is called the complement of A where U is a universal set .

  $\bar{A} = A^c = \{x \in U \mid x \notin A\}.$



•The sets whose intersection is the empty set is said to be **disjoint (mutually exclusive)** sets. If $A \cap B = \emptyset$, then A and B are disjoint sets.

Theorem: If A and B are any sets, then (A – B) and B are disjoint.
A collection of sets $\{A_1, A_2, \ldots, A_n\}$ is called *mutually* or *pairwise disjoint* if $A_i \cap A_j = \varnothing$ whenever $i \neq j$.

Cartesian Products

- Given two sets, A and B, we define the *Cartesian Product*, $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$.
- The element (a, b) is called an *ordered pair*, since (a, b) and (b, a) are distinct if $a \neq b$.
- If $A = \{1, 2, 3\}$ and $B = \{8, 9\}$, then:
  $A \times B = \{(1, 8), (1, 9), (2, 8), (2, 9), (3, 8), (3, 9)\}$
  $B \times A = \{(8, 1), (8, 2), (8, 3), (9, 1), (9, 2), (9, 3)\}$

Properties of Sets

- **Theorem** 1 (*Some subset relations*):
  1) $A \cap B \subseteq A$
  2) $A \subseteq A \cup B$
  3) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

- Commutative Laws:

$$(a)\ A \cap B = B \cap A$$

$$(b)\ A \cup B = B \cup A$$

- Associative Laws:

$$(a)\ (A \cap B) \cap C = A \cap (B \cap C)$$

$$(b)\ (A \cup B) \cup C = A \cup (B \cup C)$$

- Distributive Laws:

$$(a)\ A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(b)\ A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

- Double Complement Law:

$$(A^c)^c = A$$

- De Morgan's Laws:

$$(a)\ (A \cap B)^c = A^c \cup B^c$$

$$(b)\ (A \cup B)^c = A^c \cap B^c$$

- Absorption Laws:

$$(a)\ A \cup (A \cap B) = A$$

$$(b)\ A \cap (A \cup B) = A$$

Showing that an alleged set property is false

- **Statement:** For all sets A,B and C,

$$A-(B-C) \neq (A-B)-C \,.$$

Example:

Let $A=\{1,2,3,4\}$, $B=\{3,4,5,6\}$, $C=\{3\}$.

Then $B-C = \{4,5,6\}$ *and* $A-(B-C) = \{1,2,3\}$.

On the other hand,

$$A-B = \{1,2\} \quad \textit{and} \quad (A-B)-C = \{1,2\} \,.$$

Thus, for this example $A-(B-C) \neq (A-B)-C$.

SETS ALGEBRA RULES

	Union	Intersection
a) Identity	$A \cup \varnothing = A$ $A \cup U = U$	$A \cap \varnothing = \varnothing$ $A \cap U = A$
b) Complement	$A \cup A^C = U$ $((A)^C)^C = A$	$A \cap A^C = \varnothing$ $U^C = \varnothing, \ \varnothing^C = U$
c) Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
d) Idempotent	$A \cup A = A$	$A \cap A = A$
e) Associative	$(A \cup B) \cup C = A \cup (B \cup C);$ $(A \cap B) \cap C = A \cap (B \cap C)$	
f) Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C);$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	
g) De Morgan	$(A \cup B)^C = A^C \cap B^C$	$(A \cap B)^C = A^C \cup B^C$

If n(A) represents the number of elements in A, then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Counting means determining all the possible ways the elements of a set can be arranged.

Definition: A set with finite elements is called a finite set. Otherwise, it is said to be an infinite set. If there is a one-to-one correspondence between elements of a set and the set of natural numbers $N = \{0, 1, 2, ...\}$, then that set is called countably infinite. A set that is not countable is said to be an uncountable set.

$$A = \{x : x \text{ represents days of the week}\} \Rightarrow \text{A finite set}$$

$$B = \{x : x \text{ represents towns in the world}\} \Rightarrow \text{A finite set}$$

$$C = \{x : x \in Z \text{ (Z is set of integers}\} \Rightarrow \text{A countably infinite set}$$

$$D = \{x : 0 \le x \le 1,\ \ x \in R\} \Rightarrow \text{An uncountably set}$$

COUNTING TECHNIQUES

1) Multiplication Principle of Counting

If a task consists of a sequence of choices in which there are p selections for the first choice, q selections for the second choice, r selections for the third choice, and so on, then the task of making these selections can be done in $p \times q \times r \times \ldots$

different ways.

If we have 6 different shirts, 4 different pants, 5 different pairs of socks and 3 different pairs of shoes, how many different outcomes could we wear?

$$6 \times 4 \times 5 \times 3 = 360$$

Example: An experiment consists of tossing a coin and then rolling a die. How many possible outcomes of this experiment?

Let $A = \{H, T\}$ represents outcomes of tossing a coin where H is said head and T is tail and

$B = \{1, 2, 3, 4, 5, 6\}$ represents outcomes of rolling a die.

$$AxB = \begin{Bmatrix} (H,1), (H,2), (H,3), (H,4), (H,5), (H,6), \\ (T,1), (T,2), (T,3), (T,4), (T,5), (T,6) \end{Bmatrix}$$

$n(A)xn(B) = 12$ possible outcomes

Example: If a coin tossed successively 3 times, how many possible outcomes?

$A = \{H, T\} \longrightarrow 2x2x2 = 2^3 = 8$

We construct tree diagrams to illustrate possible outcomes of an experiment.

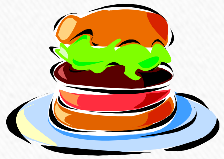
Example: For dinner you have the following choices:

ENTREES


soup salad

MAINS





chicken prawns hamburger

DESSERTS


icecream cake

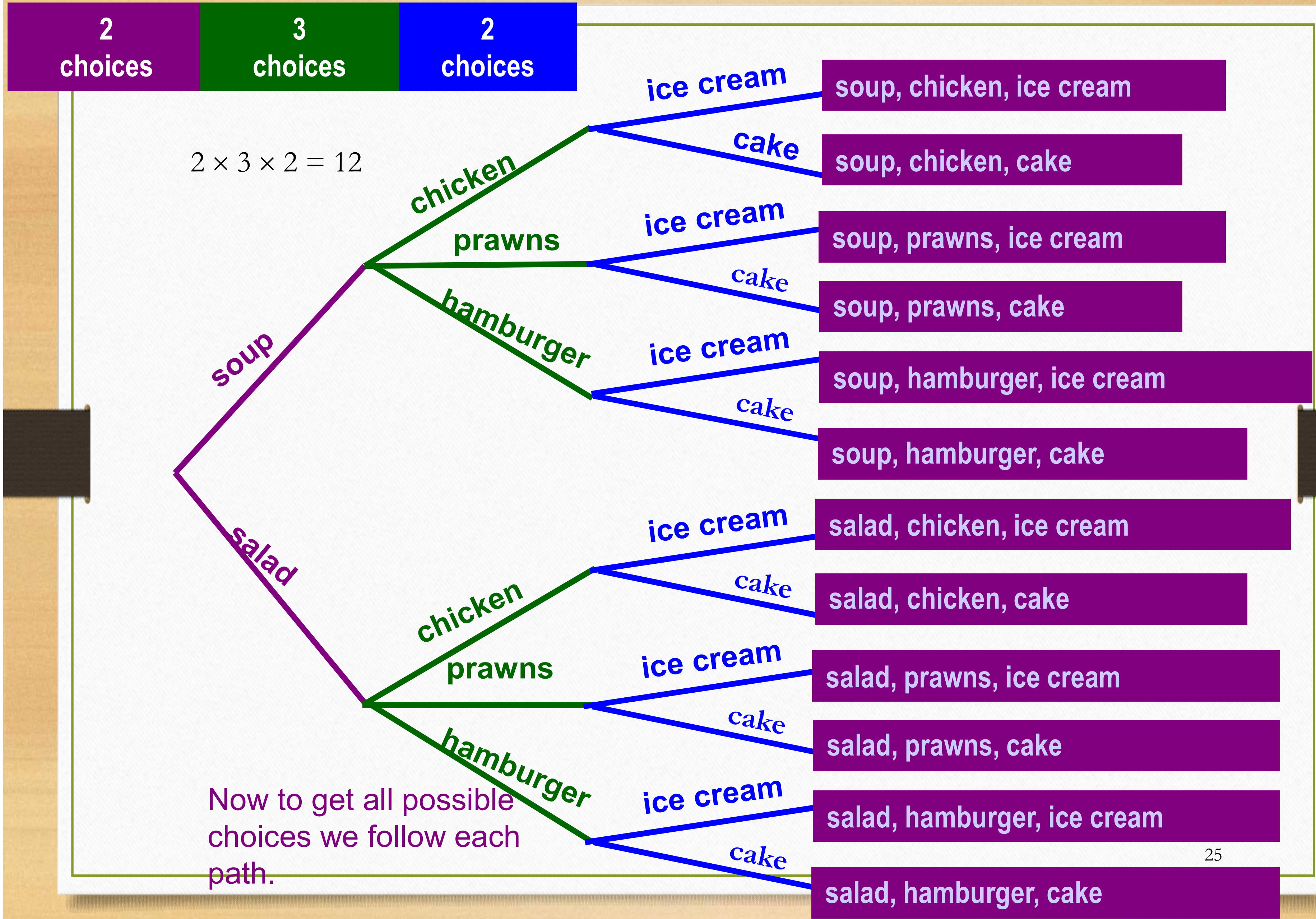
How many different combinations of meals could you make?

We'll build a tree diagram to show all of the choices.

Notice the number of choices at each branch

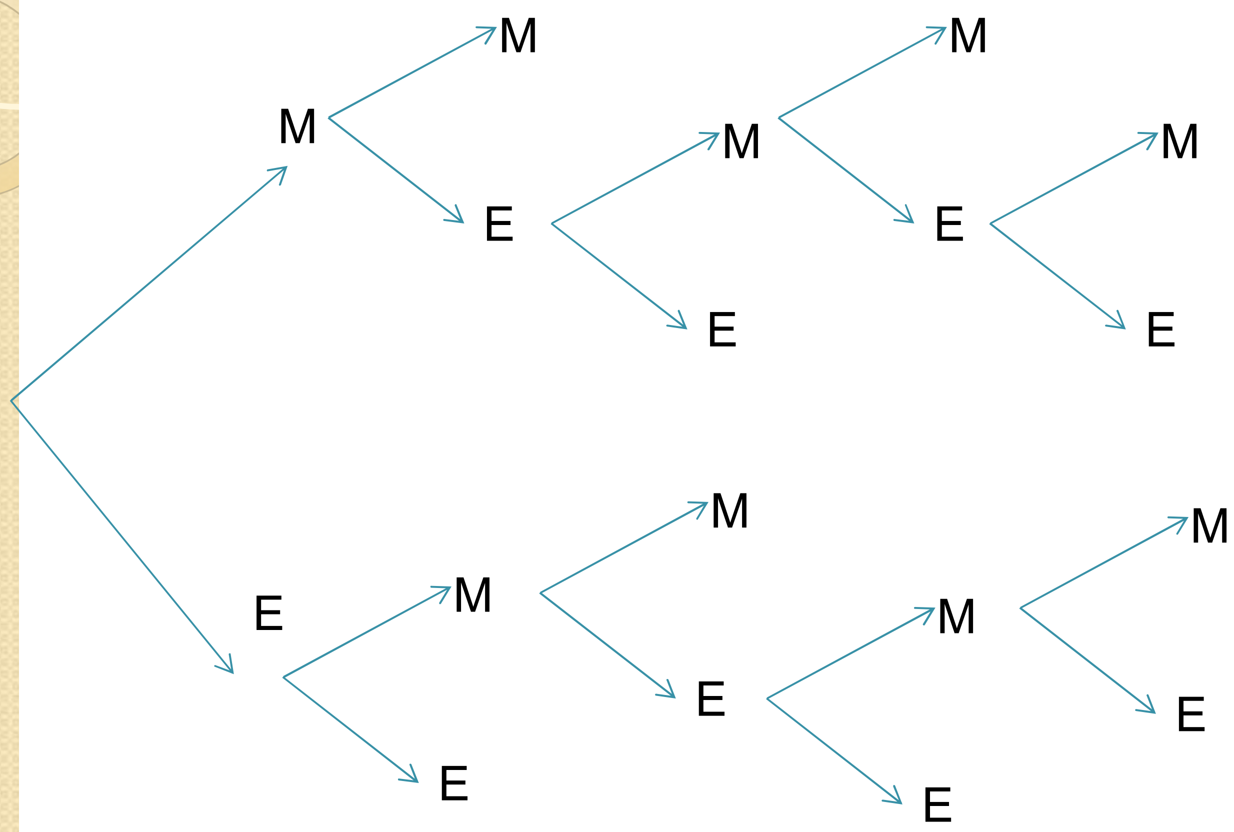
We ended up with 12 possibilities

$$2 \times 3 \times 2 = 12$$



Now to get all possible choices we follow each path.

Example: Mehmet and Emin are playing tennis. If a player gains two games successively or three games in total he wins the tournament. Find possible results by using a tree diagram.



$$\{MM, MEE, MEMM, MEMEM, MEMEE, EE, EMM, EMEE, EMEMM, EMEME\}$$

Example: A coin is tossed until a tail occurs or a head occurs four times successively. Show possible outcomes by a tree diagram.

2. PERMUTATIONS:

In general, if **r** objects are selected from a set of **n** distinct objects, any specific order of these **r** objects is called a permutation. (A **permutation** is an ordered arrangement of ***r*** objects chosen from ***n*** objects.)

i) There is no repetition

ii) Order is important

Theorem: The number of permutations denoted by P(n,r) and is calculated as;

$$\boxed{P(n,r)=\frac{n!}{(n-r)!}}$$

Note: n!=n(n-1)(n-2)….3.2.1
0!=1

$$P(n,n)=\frac{n!}{(n-n)!}=\frac{n!}{0!}=n!, \qquad P(n,0)=\frac{n!}{(n-0)!}=\frac{n!}{n!}=1$$

Note:

* The number of permutations of n distinct objects is **n!**.

* The number of permutations of n distinct objects arranged in a circle is **(n-1)!**

Example: In how many ways can 48 members of a union choose a president, a vice-president and a secretary?

$$\underbrace{48}_{P}.\underbrace{47}_{V.P}.\underbrace{46}_{S} \Rightarrow P(48,3) = \frac{48!}{(48-3)!} = 48x47x46$$

EXAMPLE

In how many different ways can one answer all the questions of a true–false test consisting of 20 questions?

Solution

Altogether there are

$$2 \cdot 2 \cdot 2 \cdot 2 \cdot \ldots \cdot 2 \cdot 2 = 2^{20} = 1{,}048{,}576$$

different ways in which one can answer all the questions; only one of these corresponds to the case where all the questions are correct and only one corresponds to the case where all the answers are wrong.

EXAMPLE

In how many different ways can the five starting players of a basketball team be introduced to the public?

Solution

There are $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways in which they can be introduced.

EXAMPLE

How many circular permutations are there of four persons playing bridge?

Solution

If we arbitrarily consider the position of one of the four players as fixed, we can seat (arrange) the other three players in $3! = 6$ different ways. In other words, there are six different circular permutations.

EXAMPLE

How many different permutations are there of the letters in the word "book"?

Solution

If we distinguish for the moment between the two o's by labeling them o_1 and o_2, there are $4! = 24$ different permutations of the symbols b, o_1, o_2, and k. However, if we drop the subscripts, then bo_1ko_2 and bo_2ko_1, for instance, both yield $boko$, and since each pair of permutations with subscripts yields but one arrangement without subscripts, the total number of arrangements of the letters in the word "book" is $\frac{24}{2} = 12$.

3. PERMUTATION WITH REPETITION (PERMUTATIONS OF MULTISETS):

Definition: The number of permutations of n objects of which

n_1 are of one kind,

n_2 of a second kind,

…..,

n_k of a k^{th} kind,

and

$n_1 + n_2 + \ldots\ldots n_k = n$

is

$$\boxed{\binom{n}{n_1, n_2, \ldots. n_k} = \frac{n!}{n_1! n_2! \ldots\ldots n_k!} = P(n; n_1, n_2 \ldots.., n_k)}$$

Example: Consider permutations of 3 Black and 2 White Balls. How many order can be obtained?

If we classify the balls as B_1,B_2,B_3 and W_1,W_2 (for each permutations), 3!2! permutations occurs. Thus, one can write;

The number of repetitive permutations x 3!2!=5!

The number of repetitive permutations= $\boxed{\frac{5!}{2!3!} = 10}$

WWBBB,WBWBB,WBBWB, WBBBW, BWWBB,
BWBWB,BWBBW, BBWWB, BBWBW,BBBWW

Example: How many different orders can be obtained from 4 red flags, 3 white flags and 1 blue flag?

$$\binom{8}{4,3,1} = \frac{8!}{4!3!1!} = \frac{8.7.6.5.4!}{4!3!1!} = 280$$

4. COMBINATIONS:

If **r** objects are selected from a set of **n** distinct objects, a selection set without regard to their order is called a combination. The number of combinations is denoted by C(n,r).

i) No repetition
ii) Order is not important

$$\boxed{C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}}$$

Note: The difference between permutation and combination is:
For combination order is not important but for permutation it is important.

$$C(n,r) = \binom{n}{r} = \frac{P(n,r)}{r!}$$

Example: Consider 24 permutations of three of alphabet's first four letters as given below.

$$\{a,b,c\} \rightarrow abc, acb, bac, bca, cab, cba \qquad 6 = 3!$$

$$\{a,b,d\} \rightarrow abd, adb, bad, bda, dab, dba \qquad 6 = 3!$$

$$\{a,c,d\} \rightarrow .... \qquad 6 = 3!$$

$$\{b,c,d\} \rightarrow .... \qquad 6 = 3!$$

$$P(4,3) = 24 = \text{The number of combinations x } 3!$$

$$C\binom{n}{r} = \frac{P(4,3)}{3!} = \frac{4!}{(4-3)!3!} = 4$$

Example: In how many ways can a person choose 3 flags from a set of 5 distinct flags, assuming that the order in which the flags are chosen is of no consequence?

$$C\binom{5}{3} = \frac{5!}{(5-3)!3!} = 10$$

Example:

In how many different ways can six tosses of a coin yield two heads and four tails?

Solution

This question is the same as asking for the number of ways in which we can select the two tosses on which heads is to occur.

$$\binom{6}{2} = \frac{6!}{2! \cdot 4!} = 15$$

This result could also have been obtained by the rather tedious process of enumerating the various possibilities, HHTTTT, TTHTHT, HTHTTT, . . . , where H stands for head and T for tail.

Example:

How many different committees of two chemists and one physicist can be formed from the four chemists and three physicists on the faculty of a small college?

Solution

Since two of four chemists can be selected in $\binom{4}{2} = \frac{4!}{2! \cdot 2!} = 6$ ways and one of three physicists can be selected in $\binom{3}{1} = \frac{3!}{1! \cdot 2!} = 3$ ways the number of committees is $6 \cdot 3 = 18$.

$\begin{pmatrix} n \\ r \end{pmatrix}$ is also said to be **binomial coefficient**.

*If it is binomial coefficient, then it can be defined for negative and real numbers.

$$\boxed{\begin{pmatrix} n \\ r \end{pmatrix} = \frac{n!}{r!(n-r)!} = \frac{\overbrace{n(n-1)(n-2)\ldots(n-r+1)}^{\text{r terms}}}{r!}}$$

Example:

$$\begin{pmatrix} 8 \\ 5 \end{pmatrix} = \frac{8(8-1)(8-2)\ldots(8-5+1)}{5!} = \frac{8.7.6.5.4}{5!}$$

$$\begin{pmatrix} -8 \\ 5 \end{pmatrix} = \frac{-8(-8-1)(-8-2)\ldots(-8-5+1)}{5!} = -\frac{8.9.10.11.12}{5!} = -\begin{pmatrix} 12 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 1/2 \\ 3 \end{pmatrix} = \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!} = \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}$$

If n>0,

1) C(n,n)=C(n,0)=C(0,0)=1,
2) C(n,1)=n
3) C(n,r)=C(n,n-r)
4) C(n,r)=0, for r>n ($n \in N$) or r<0

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Example: Show that $\boxed{\binom{-r}{k} = (-1)^k \binom{r+k-1}{k}}$

Solution:

$$\binom{-r}{k} = \frac{(-r)(-r-1)(-r-2).....(-r-k+1)}{k!}$$

$$= (-1)^k \frac{(r)(r+1)(r+2).....(r+k-1)}{k!} \frac{(r-1)!}{(r-1)!}$$

$$= (-1)^k \frac{(r+k-1)!}{k!(r-1)!} = (-1)^k \binom{r+k-1}{k}$$

Pascal's Rule

$$\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r} \quad \text{for } 1 \le r \le n$$

Proof:

$$\frac{n!}{(r-1)!\,(n-r+1)!} + \frac{n!}{r!\,(n-r)!}$$

$$(r) \qquad (n-r+1)$$

$$\frac{n!\,[\cancel{r} + (n-\cancel{r}+1)]}{r!\,(n-r+1)!} = \frac{(n+1)!}{r!\,(n-r+1)!} = \binom{n+1}{r}$$

Example: There are six candidates representing the contest. Since a voter can use the vote by marking the names of one or two candidates, how many different ways can the voter use the vote?

Solution: There are 2 discrete cases. In other words, voting can be done for 1 candidate or 2 candidates. If the voter votes for 1 candidate, $\boxed{\binom{6}{1}}$ different ways occur, if the voter votes for 2 candidates $\boxed{\binom{6}{2}}$ different ways occur. Therefore, according to the collection rule and Pascal rule

$$\boxed{\binom{6}{1}+\binom{6}{2}=\binom{7}{2}}$$

vote in different ways.

5. COMBINATIONS WITH REPETITION:

Definition: Consider a group containing n distinct objects. If we select an item k times from this group with replacing them back, each selection (with k objects) without regard to their order is called a combination with repetititon.

i) Order is not important

ii) There is repetition

The number of these selections can be calculated by

$$\boxed{\binom{n+k-1}{k} = \binom{n+k-1}{n-1}}$$

Example: Find the numbers of sixth order partial derivatives of the function $f(x, y, z) = x^3 e^{yz} + (xy \sin z)^{10}$

Solution: Here we choose 6 objects in $\{x, y, z\}$

n=3

k=6

$$\binom{3+6-1}{6} = \binom{8}{6} = 28$$

Possible combinations $\Rightarrow$ f_{xxxxxx}, f_{xxxxxy}, f_{xxxxyy}, ………, f_{zzzzzz}

6. BINOMIAL EXPANSION:

If **n** is a natural number and **a** and **b** are real numbers then

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \ldots.. + \binom{n}{n} a^0 b^n$$

$$= a^n + na^{n-1}b + \frac{n(n-1)}{1.2} a^{n-2} b^2 + \ldots\ldots. + nab^{n-1} + b^n$$

$\Downarrow$

Binomial Coefficient

There is a relationship between binomial expansion and Pascal's Triangle, such as

$$\begin{array}{lcccccccccc}
n=0 & \Rightarrow & & & & & 1 & & & & \\
n=1 & \Rightarrow & & & & 1 & & 1 & & & \\
n=2 & \Rightarrow & & & 1 & & 2 & & 1 & & \\
n=3 & \Rightarrow & & 1 & & 3 & & 3 & & 1 & \\
n=4 & \Rightarrow & 1 & & 4 & & 6 & & 4 & & 1
\end{array}$$

$$\binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4}$$

$(a+b)^n$ expansion has the following characteristics.

i) There are n + 1 terms.
ii) The sum of the exponents of a and b in each term is n.
iii) The exponents of a are reduced from n to 0, and the exponents of b from 0 to n.
iv) Coefficient of the term $a^t b^s$ in the expansion $(a+b)^n$ where t + s = n is;

$$\binom{n}{t} = \binom{n}{s}$$

v) Coefficients of terms equal to the ends are equal, because of the relation;

$$\binom{n}{r} = \binom{n}{n-r}$$

Example: Show that

a) $$\boxed{\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\ldots+\binom{n}{n}=2^n}$$

$$(a+b)^n=\sum_{r=0}^{n}\binom{n}{r}a^{n-r}\cdot b^r=\binom{n}{0}a^n b^0+\binom{n}{1}a^{n-1}\cdot b^1+\ldots+\binom{n}{n}a^0 b^n$$

$a=b=1$

$$2^n=(1+1)^n=\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\ldots+\binom{n}{n}$$

b) $$\boxed{\binom{n}{0}-\binom{n}{1}+\binom{n}{2}+\ldots(-1)^n\binom{n}{n}=0}$$

$a=-b=1$

$$(1+(-1))^n=(1-1)^n=0=\binom{n}{0}-\binom{n}{1}+\binom{n}{2}-\ldots+(-1)^n\cdot\binom{n}{n}$$

Note: If n is **not a natural number**, then, the expansion will not be finite, it will contain an infinite number of terms.

$$\boxed{(a+b)^n = \sum_{r=0}^{\infty} \binom{n}{r} a^{n-r} b^r}$$

$$\boxed{\binom{-r}{k} = (-1)^k \binom{r+k-1}{k}}$$

This expansion converges for $|b| < |a|$

Example:

$$(1+x)^{-5} = \sum_{r=0}^{\infty} \binom{-5}{r} 1^{-5-r} x^r = \binom{-5}{0} 1^{-5-0} x^0 + \binom{-5}{1} 1^{-5-1} x^1 + \binom{-5}{2} 1^{-5-2} x^2 + \ldots..$$

$$= \binom{4}{0} = 1 + (-1)^1 \binom{5+1-1}{1} x + (-1)^2 \binom{5+2-1}{2} x^2 + \ldots..$$

$$= 1 - \binom{5}{1} x + (-1)^2 \binom{6}{2} x^2 + \ldots..$$

$$= \sum_{i=0}^{\infty} (-1)^i \binom{4+i}{i} x^i$$ If $|x| < 1$ it converges.

$$\binom{-5}{0} = (-1)^0 \binom{5+0-1}{0} = 1 \cdot \binom{4}{0}$$

7. MULTINOMINAL EXPANSION:

Definition: If n is **a natural number**, then,

$$\boxed{(x_1 + x_2 + \ldots\ldots + x_k)^n = \sum_{n_1, n_2, \ldots\ldots, n_k} \binom{n}{n_1, n_2, \ldots\ldots n_k} x_1^{n_1} x_2^{n_2} \ldots\ldots x_k^{n_k}}$$

where $n_1, n_2, \ldots\ldots n_k$ are the natural numbers and $n_1 + n_2 + \ldots\ldots + n_k = n$

$$\binom{n}{n_1, n_2, \ldots\ldots n_k} = \frac{n!}{n_1! n_2! \ldots\ldots n_k!}$$

Number of terms in multinominal expansion including $(x_1 + x_2 + \ldots\ldots + x_k)^n$

is $\boxed{\binom{n+k-1}{n}}$

Example:

$$(x_1+x_2+x_3)^4 = \sum_{\substack{n_1,n_2,n_3 \\ \left(\begin{array}{l} n_1+n_2+n_3=4 \\ n_j \in N \end{array}\right)}} \frac{4!}{n_1!n_2!n_3!} x_1^{n_1}x_2^{n_2}x_3^{n_3}$$

The triples; (n_1,n_2,n_3) for $n_1+n_2+n_3=4$ and n_j are natural numbers are

$(4,0,0)$, $(3,1,0)$, $(3,0,1)$, $(2,2,0)$, $(2,1,1)$, $(2,0,2)$, $(1,3,0)$, $(1,0,3)$,

$(1,2,1)$, $(1,1,2)$, $(0,4,0)$, $(0,0,4)$, $(0,3,1)$, $(0,1,3)$, $(0,2,2)$

$$(x_1+x_2+x_3)^4 = \frac{4!}{4!}x_1^4 + \frac{4!}{3!}x_1^3x_2 + \frac{4!}{3!}x_1^3x_3 + \ldots\ldots + \frac{4!}{4!}x_3^4$$

$$= x_1^4 + 4x_1^3x_2 + 4x_1^3x_3 + \ldots\ldots + x_3^4$$

The number of terms from equation is;

$$\boxed{\binom{n+k-1}{n} = \binom{4+3-1}{4} = \binom{6}{4} = 15}$$

EXAMPLE

What is the coefficient of $x_1^3 x_2 x_3^2$ in the expansion of $(x_1 + x_2 + x_3)^6$?

Solution

Substituting $n = 6, r_1 = 3, r_2 = 1$, and $r_3 = 2$ into the preceding formula, we get

$$\frac{6!}{3! \cdot 1! \cdot 2!} = 60$$

EXAMPLES:

1. In how many ways can two single women and three married couples be seated around a circular table if

   $3 \times 2 = 6$

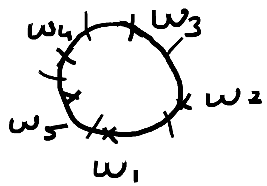
   a. there is no condition,
   b. two single women must not sit next to each other,
   c. men must not sit next to each other.

2. How many words can be formed using all the letters of "KARAÇAKÖK" so that the letters 'A' never come together?

① a) $(8-1)! = 7!$

b) $(7-1)! \cdot 2!$

$7! - 6!\,2!$

c) 

$(5-1)!$

$\binom{5}{3}\,3! = \frac{5!}{\cancel{3!}\cdot 2!}\,\cancel{3!}$

$= \frac{5\cdot 4\cdot 3\,\cancel{2!}}{\cancel{2!}}$

$= 60$

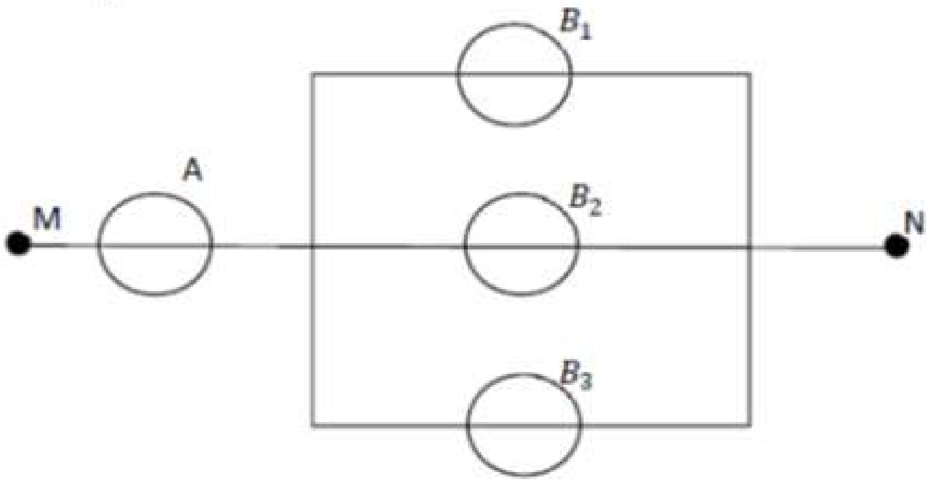
$4! \cdot 60$

② $\frac{6!}{3!\cdot 1!\cdot 1!}$

$\binom{7}{3}$

_ _ _ _ _ _ $\Big\}$ $\frac{6!}{3!\cdot 1!\cdot 1!} \cdot \binom{7}{3}$

3. The figure given below shows electrical circuit between points M and N. The event A and B are defined as corruption of the device A and device $B_k, k = 1,2,3$, respectively. If the event C is defined as cutting of electric current between points M and N, determine the event C and $\bar{C}$ in terms of A and B_k.



$$\bar{C} = \bar{A} \cap (\bar{B}_1 \cup \bar{B}_2 \cup \bar{B}_3)$$

$$C = \left[\bar{A} \cap (\bar{B}_1 \cup \bar{B}_2 \cup \bar{B}_3)\right]^c = A \cup (B_1 \cap B_2 \cap B_3)$$

How many different ways are placed 7 balls in 7 boxes, 2 in two of the boxes, 1 in three of the boxes and two are empty?

7 boxes are divided into 3 classes according to this dist.

In the first class 2 boxes with 2 balls.
" second " 3 " " 1 "
" third " 2 " are empty

$$\frac{7!}{2!\cdot 3!\cdot 2!} = 210$$

The number of different permutations of balls

$$\frac{7!}{2!\cdot 2!\cdot 1!\cdot 1!\cdot 1!\cdot 0!\cdot 0!} = \frac{7!}{2!\cdot 2!} = 1260$$

210×1260 //

There is an engine and 2 boilers in one installation. The event A is defined the engine working well, and the events B_k $(k=1,2)$ are defined the k-th boilers working well. And the event C is defined inst. works. Determine the event C and $\bar{C}$ in terms of A and B_k.

(If the engine and at least one boiler is working well, the ins. works.)

$A = \{$ Engine is working well $\}$

$B_k = \{$ k-th boiler is " " $\}$

$C = \{$ Installation process $\}$

$$C = A \cap (B_1 \cup B_2)$$

$$\bar{C} = \bar{A} \cup (\bar{B}_1 \cap \bar{B}_2)$$

How many diffent Commissions are selected from 4 boys and 2 girls, with the condition at least 1 girl and the number of boys not less than 2 times of the number of girls?

* If 1 girl is selected then the number of boys can be 2, 3 or 4

$$\binom{2}{1}\left[\binom{4}{2}+\binom{4}{3}+\binom{4}{4}\right]$$

* If 2 girls are selected the - - - - - $\Rightarrow$ 4

$$\binom{2}{2}\binom{4}{4}$$

$$\binom{2}{1}\left[\binom{4}{2}+\binom{4}{3}+\binom{4}{4}\right]+\binom{2}{2}\binom{4}{4}$$

$$(X \cup A)^c \cup (X \cup A^c)^c = B \qquad X = ?$$

$$(X \cup A)^c = X^c \cap A^c$$

$$(X \cup A^c)^c = X^c \cap A$$

$$(X^c \cap A^c) \cup (X^c \cap A) = X^c \cap \underbrace{(A^c \cup A)}_{U,\,E} = B$$

$$\Rightarrow X^c = B$$

$$(X^c)^c = B^c$$

$$X = B^c = \overline{B}$$